
Slide 1 – Title Slide

AI-Generated Credit Scoring using Alternative Data
Building Fair, Explainable & Scalable Credit Risk Models

Team Name: XYZ

Problem Domain: Financial Risk & Inclusion

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Slide 2 – Problem Statement

- Millions of individuals and MSMEs lack traditional credit histories
- Existing credit scoring models:
- Are rigid
- Ignore alternative data
- Exclude underbanked populations
- Manual underwriting is:
- Expensive
- Slow
- Inconsistent

Goal:

Design an AI-powered credit scoring system that predicts default risk using alternative data, while ensuring fairness, explainability, and regulatory compliance.

Slide 3 – Business Impact

- Expand customer base (unbanked & thin-file users)
 - Improve default risk prediction ($AUC > 0.75$)
 - Reduce cost per credit decision
 - Enable transparent & explainable lending decisions
 - Support financial inclusion at scale
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Slide 4 – Solution Overview

Our solution is a machine-learning driven credit scoring pipeline that:

- Uses structured financial & behavioral data
- Applies multiple ML models for comparison
- Handles class imbalance
- Provides explainability using SHAP
- Aligns with RBI & EU AI Act guidelines

Deliverable:

A single, modular Jupyter Notebook ML pipeline

Slide 5 – System Architecture

Linear ML Pipeline (Single Notebook)

```
code

Data Loading
  ↓
Exploratory Data Analysis
  ↓
Preprocessing & Feature Engineering
  ↓
Model Training (Baseline → Advanced)
  ↓
Evaluation & Metrics
  ↓
Explainability (SHAP)
  ↓
Final Results Summary
```

Slide 6 – Data Sources

Public & Anonymized Datasets

Public & Anonymized Datasets

Dataset	Purpose
German Credit (Statlog)	Rapid prototyping
Give Me Some Credit	Medium-scale validation
Home Credit Default Risk	Final large-scale evaluation

Why these?

- Real-world credit behavior
 - Class imbalance (realistic)
 - Regulatory-safe & public
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Slide 7 – Key Features Used

- Credit amount
- Credit history length
- Loan purpose
- Repayment behavior
- Employment stability
- Account balances
- Previous defaults

⚠ Protected attributes (gender, religion, ethnicity) excluded

Slide 8 – Data Challenges & Handling

Challenges

- Class imbalance (~5–10% defaults)
- Missing values
- Mixed categorical & numerical data

Solutions

- Median/mode imputation
 - One-hot encoding
 - SMOTE for class imbalance
 - Stratified train-test split
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Slide 9 – Models Implemented

We implemented three models:

We implemented **three models**:

Model	Role
Logistic Regression	Baseline & interpretability
Random Forest	Feature importance
XGBoost	Final production model

Slide 10 – Why These Models?

- Logistic Regression
 - Simple, interpretable
 - Industry baseline
 - Random Forest
 - Handles non-linearity
 - Robust to noise
 - XGBoost
 - Best performance on tabular data
 - Handles imbalance well
 - Industry-grade (used by banks)
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Slide 11 – Model Evaluation Metrics

We focus on threshold-independent & imbalance-aware metrics:

- AUC-ROC (primary)
- Precision
- Recall
- F1-Score
- Confusion Matrix

⚠ Accuracy alone is misleading in credit data

Slide 12 – Results Summary

Model	AUC-ROC
Logistic Regression	~0.71
Random Forest	~0.75
XGBoost	~0.80+
✅ XGBoost selected as final model	

Slide 13 – Visual Evaluation

- ROC Curve comparison
- Confusion Matrix
- Precision-Recall tradeoff

(Show screenshots from notebook)

Slide 14 – Explainability (SHAP)

Credit decisions must be explainable.

We use SHAP (SHapley Additive exPlanations) to:

- Explain global feature importance
- Justify individual loan decisions

Slide 15 – SHAP Visuals

- SHAP Summary Plot (global)
- SHAP Force Plot (individual applicant)

- ✓ Meets RBI Fair Practices Code
- ✓ Aligns with EU AI Act requirements

Slide 16 – Fairness & Bias Considerations

- Removed protected attributes
- Audited performance across subgroups
- Evaluated disparate impact
- Ensured no disproportionate rejection

Outcome:

Model balances risk control + inclusion

Slide 17 – Deployment Strategy

- Google Colab / Kaggle Notebook
- No external infrastructure required
- Easily portable to:
- API
- Batch scoring
- Bank risk systems

Slide 18 – Benefits

- Faster credit decisions
- Lower underwriting cost
- Expanded customer reach

- Transparent & auditable AI
 - Scalable across emerging markets
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Slide 19 – Future Scope

- Add real alternative data (utility, telecom)
 - Time-series credit behavior modeling
 - Deep learning for transactional data
 - Live API deployment
 - Continuous bias monitoring
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Slide 20 – Conclusion

- Built an end-to-end AI credit scoring system
- Achieved strong predictive performance
- Ensured fairness & explainability
- Ready for real-world deployment

AI + Responsible Lending = Financial Inclusion